INVESTMENTS

Ninth Canadian Edition · Bodie, Kane, Marcus

CHAPTER 8

Index Models

Verified against source text — learning objectives, formulas, and worked examples checked directly against the chapter.

Study Guide · Prepared for Personal Use

LO1 · A Single-Factor Security Market

The Markowitz model (Chapter 7) requires a huge number of covariance estimates and gives no guidance on forecasting expected returns. **Index models** fix both problems by assuming security returns are driven by one common macroeconomic factor plus firm-specific noise.

The Single-Factor Model

Each security's return is decomposed into three pieces: its originally expected return, the impact of a common macroeconomic surprise (scaled by the security's sensitivity, beta), and a firm-specific surprise. Securities with higher **beta** (e.g., cyclical firms) respond more strongly to macro shocks and carry more **systematic risk.** Because firm-specific surprises are assumed uncorrelated across firms, the **only** source of covariance between any two securities is their shared dependence on the common factor — which is what makes the model so much simpler to estimate than the full Markowitz covariance matrix.

> *Total risk splits cleanly into two uncorrelated pieces: systematic risk (driven by beta and market variance) and firm-specific risk (the variance of the residual). This decomposition is the single most important idea in the chapter.*

LO2 · The Single-Index Model

To make the single-factor model usable, we need an observable proxy for the macro factor — typically a broad market index (e.g., the S&P/TSX Composite). We regress each security's **excess** return ($r_i - r_f$) on the index's excess return (see Formulas page, Equation 8.8). We use excess rather than total returns because it is the *spread* between the market return and the risk-free rate — not the market return's absolute level — that signals good or bad macroeconomic news (an 8% stock return was disappointing when bills yielded 10%, but excellent when bills yielded 3%).

Alpha, Beta, and the Security Characteristic Line (SCL)

- **Alpha (α) —** the intercept: a security's expected excess return when the market's excess return is zero. Represents a non-market return component — e.g., mispricing an analyst believes they've identified.
- **Beta (β) —** the slope: the security's sensitivity to the index. A 1% move in the index moves the security's expected excess return by β%.
- **Residual (e) —** the zero-mean firm-specific surprise, i.e., the vertical distance of each observation from the fitted regression line (the security characteristic line, or SCL).

Taking expectations of the regression equation gives the **expected return–beta relationship:** a security's risk premium equals its alpha (non-market compensation) plus its beta times the market risk premium (systematic compensation). If markets were in full equilibrium, competition would drive alpha to zero for every security — the deeper implication explored in Chapter 9's CAPM.

LO3 · Estimating the Single-Index Model

The textbook estimates the model using 60 months of data (2013–2017) for six large Canadian stocks — BCE (telecom), CNR (industrials), Enbridge and Suncor (energy), and Royal Bank and TD Bank (financials) — regressed against the S&P/TSX Composite.

Reading the Regression Output

- **Correlation —** measures how closely a stock tracks the index. A moderate correlation (e.g., 0.45) means the stock follows the market's general direction but with substantial independent movement.
- **R-square (coefficient of determination) —** the percentage of a security's return variance explained by the market factor; equivalently, 1 minus the ratio of residual variance to total variance. The remainder is firm-specific risk. Across the six Canadian stocks studied, the average R-square was only 21.4% — substantially lower than for a comparable sample of U.S. stocks (where firm-specific risk was 70.5% of the total, versus 78.6% for the Canadian sample), suggesting Canadian large-caps carry even more idiosyncratic risk relative to the market.
- **Standard error & t-statistic —** the standard error measures estimation imprecision; the t-statistic (coefficient ÷ standard error) tests whether the true coefficient could plausibly be zero. A t-statistic around 2 or higher (roughly a 5% significance level) is conventionally considered statistically significant.

A Crucial Caveat on Alpha

Even when an estimated alpha is economically meaningful, it is usually **not** statistically significant in a single 5-year sample — and, more importantly, extensive empirical evidence shows historical alpha estimates **do not persist** from one period to the next. A stock's past alpha is a poor forecast of its future alpha, which is precisely why security analysis (and market efficiency, covered in Chapter 11) is so difficult in practice.

LO4 · The Industry Version of the Index Model

Commercial index-model services (e.g., Bloomberg, Value Line) typically estimate beta using **total** returns rather than excess returns, since at short (e.g., daily) intervals the risk-free rate is close enough to zero that the distinction barely matters in practice. They also often report a **adjusted ("shrunk") beta** — a weighted average of the sample beta and 1.0 — reflecting the empirical tendency of estimated betas to drift toward the market average of 1.0 over time.

Beta Capture and Tracking Portfolios

Index-model regressions are also the engine behind **tracking portfolios:** a portfolio engineered to have a specific beta (often zero, "beta capture") relative to the index, isolating a fund’s alpha-generating bets from its exposure to overall market moves. Hedge funds commonly use this technique to strip out unwanted systematic risk and focus their bets purely on perceived firm-specific mispricing.

LO4 · Portfolio Construction Using the Single-Index Model

The index model also simplifies **optimal portfolio construction.** The optimal risky portfolio turns out to be a combination of two component portfolios: an **active portfolio (A)** built from securities with non-zero alpha (i.e., analyzed securities believed to be mispriced), and the **passive market-index portfolio (M).** This is known as the **Treynor-Black model.**

LO4 · The Information Ratio and Optimal Active Weighting

Within the active portfolio, each security is weighted in proportion to $\alpha_i / \sigma^2(e_i)$ — balancing the security's alpha contribution against the extra firm-specific variance it adds. The overall weight placed on the active portfolio versus the index is then adjusted for the active portfolio's own beta (see Formulas page).

The Information Ratio

The ratio $\alpha_A / \sigma(e_A)$ — the active portfolio's alpha divided by its residual (firm-specific) standard deviation — is called the **information ratio.** It measures how much extra Sharpe-ratio benefit an investor obtains from security analysis, per unit of extra firm-specific risk absorbed. Maximizing the *overall* Sharpe ratio of the complete risky portfolio is mathematically equivalent to maximizing the information ratio of the active portfolio.

The Payoff from Security Analysis — Squared Sharpe Ratios

The key result (see Formulas page) shows that the optimally constructed portfolio's squared Sharpe ratio equals the passive index's squared Sharpe ratio **plus** the squared information ratio of the active portfolio. Good security analysis (high alpha relative to residual risk) can only ever **add** to the passive benchmark's performance — it never subtracts, because an investor is always free to hold zero weight in a security with an unfavourable alpha-to-risk trade-off (or short it, if permitted). A negative-alpha security optimally receives a negative (short) weight in the active portfolio; if short sales are banned, it is simply dropped from consideration.

> *Beta is "neither vice nor virtue" — it simultaneously determines both a security's contribution to systematic risk and its contribution to systematic (market) risk premium, in exactly offsetting proportion. Only alpha (relative to residual risk) represents a genuine, uncompensated-for-free opportunity to improve the Sharpe ratio.*

Chapter 8 — Big-Picture Takeaways

- Index models decompose security risk into systematic (market-driven, non-diversifiable) and firm-specific (diversifiable) components, drastically reducing the inputs needed versus the full Markowitz model.
- The security characteristic line (SCL) — the regression of excess security returns on excess index returns — yields alpha (non-market return), beta (market sensitivity), and R-square (fraction of variance explained by the market).
- For a sample of large Canadian stocks, only about 21% of return variance was explained by the market factor — substantially less than the corresponding U.S. figure, implying relatively higher firm-specific risk in this sample.
- Estimated alpha values are typically statistically insignificant in any single sample and do not reliably persist across periods — a central reason security analysis is difficult.
- The Treynor-Black model splits the optimal risky portfolio into an active (alpha-driven) component and a passive index component, weighted by the information ratio ($\alpha/\sigma(e)$).

- The squared Sharpe ratio of the optimal portfolio equals the index's squared Sharpe ratio plus the active portfolio's squared information ratio — skilled security analysis can only improve on a passive benchmark, never hurt it.

The single-index regression and the Treynor-Black active-portfolio formulas are the two formula families to master here.

F1 | Single-Factor Model (Total Return)

$$r_i = E(r_i) + \beta_i \times m + e_i$$

$$\sigma_i^2 = \beta_i^2 \sigma_m^2 + \sigma^2(e_i)$$

m = the surprise in the common macro factor; e_i = firm-specific surprise (zero mean, uncorrelated across firms and with m). Systematic risk = $\beta_i^2 \sigma_m^2$; firm-specific risk = $\sigma^2(e_i)$.

F2 | Single-Index Regression (Excess Returns) — Equation 8.8

$$R_i = \alpha_i + \beta_i R_M + e_i$$

$$\text{where } R_i = r_i - r_f \quad \text{and} \quad R_M = r_M - r_f$$

Regress the security's excess return on the index's excess return. α_i = intercept (non-market return); β_i = slope (market sensitivity).

F3 | Expected Return–Beta Relationship

$$E(R_i) = \alpha_i + \beta_i E(R_M)$$

A security's risk premium splits into a non-market component (α_i) and a systematic component (β_i times the market risk premium).

F4 | R-Square (Coefficient of Determination)

$$R^2 = 1 - [\sigma^2(e_i) / \sigma_i^2] = \beta_i^2 \sigma_M^2 / \sigma_i^2$$

The fraction of a security's total return variance explained by the market index. $1 - R^2$ is the fraction due to firm-specific risk.

F5 | Optimal Sharpe Ratio — Treynor-Black Decomposition

$$S_P^2 = S_M^2 + [\alpha_A / \sigma(e_A)]^2$$

$\alpha_A/\sigma(e_A)$ is the information ratio of the active portfolio. The optimal combined portfolio's squared Sharpe ratio always ≥ the passive index's squared Sharpe ratio — security analysis can only help, never hurt, if implemented optimally.

F6 | Optimal Weight on Each Security Within the Active Portfolio

$$w_i = [\alpha_i / \sigma^2(e_i)] / \Sigma[\alpha_j / \sigma^2(e_j)]$$

Weights are proportional to each security's alpha-to-residual-variance ratio, scaled so the active-portfolio weights sum to 1. Securities with negative alpha get negative (short) weights.

EXAMPLE | The Security Characteristic Line for Suncor

Given: 60 months (2013–2017) of Suncor Energy (SU) excess returns regressed on S&P/TSX Composite excess returns.

Statistic	Estimate	Interpretation
Alpha (α)	0.0016 (0.16%/mo)	1.92% annualized; SE = 0.0064, t-stat = 0.256 — not significant
Beta (β)	1.1048	SE = 0.2856, t-stat = 3.87 — significantly different from zero
R-square	0.2052	20.5% of Suncor's return variance explained by the market
Correlation	0.4530	Moderate co-movement with the S&P/TSX Composite

Interpretation: Suncor's beta of 1.10 confirms it is slightly more volatile than the market (a "mildly aggressive" stock). Despite an economically notable annualized alpha of 1.92%, the large standard error means we cannot statistically distinguish this from a true alpha of zero — and even if it were significant within this sample, historical alpha estimates are known not to persist, so this would still be a poor basis for a forecast of Suncor's future performance.

EXAMPLE | Treynor-Black Sharpe Ratio Improvement

Given: The passive index has a Sharpe ratio $S_M = 0.40$. An analyst constructs an active portfolio with alpha $\alpha_A = 4\%$ and residual standard deviation $\sigma(e_A) = 12\%$.

Step 1 — Information ratio:

$$\alpha_A / \sigma(e_A) = 4 / 12 = 0.333$$

Step 2 — Optimal combined Sharpe ratio:

$$S_P^2 = 0.40^2 + 0.333^2 = 0.160 + 0.111 = 0.271 \quad \rightarrow \quad S_P = \sqrt{0.271} = 0.52$$

Skilled security analysis (a favourable information ratio) raises the achievable Sharpe ratio from 0.40 to 0.52 — a meaningful improvement purely from optimally blending the active portfolio with the passive index.

PRACTICE PROBLEM | Sundog Quantitative Partners

Scenario

You are a quantitative analyst at Sundog Quantitative Partners in Calgary. You have run single-index regressions (against the S&P/TSX Composite) for two Alberta-based energy stocks over the past 60 months:

Stock	Alpha (annual)	Beta	Residual SD (annual)
PrairieSky Resources	3.0%	1.20	18%
Athabasca Energy	−1.5%	0.85	22%

Additional info: The S&P/TSX Composite has an expected excess return (risk premium) of 6% and a standard deviation of 16%. The risk-free rate is 4%.

Part A — Systematic vs. Firm-Specific Risk

(i) Calculate the total variance of PrairieSky's returns using its beta, residual SD, and the market's SD.

(ii) Calculate PrairieSky's R-square. What fraction of its risk is firm-specific?

(iii) Without calculating, would you expect Athabasca Energy's R-square to be higher or lower than PrairieSky's, given the data above? Explain.

Part B — Information Ratios

(i) Calculate the information ratio for PrairieSky.

(ii) Calculate the information ratio for Athabasca Energy.

(iii) Based on information ratios alone, which stock contributes more favourably to an active portfolio's Sharpe ratio? Explain what a negative alpha implies for portfolio weighting.

Part C — Active Portfolio Construction

(i) Using the optimal weighting formula, calculate the relative weights (before scaling) that PrairieSky and Athabasca Energy would receive within the active portfolio.

(ii) Which stock receives a larger weight, and does either receive a negative (short) weight? Explain why.

Part D — Portfolio Sharpe Ratio Improvement

Suppose the resulting active portfolio (combining both stocks optimally) has an overall information ratio of 0.28. The passive index has a Sharpe ratio of 0.375 (using the market data given).

(i) Verify the passive index Sharpe ratio using the risk premium and standard deviation given.

(ii) Calculate the optimal combined portfolio's Sharpe ratio using the Treynor-Black formula.

(iii) Express in one sentence what this improvement represents in terms of the value added by Sundog's security analysis.

Part E — Practical Limits

A junior colleague suggests using these same alpha estimates, unchanged, to set next year's return forecasts for PrairieSky and Athabasca Energy.

(i) Referencing LO3, explain why this would be a questionable practice.

(ii) What statistical evidence from the chapter (e.g., regarding Suncor's alpha) supports your answer?

Hints & Guidance

Part A tests LO1 and Formulas F1 & F4. Part B tests LO4 and Formula F5's building block ($\alpha/\sigma(e)$). Part C tests LO4 and Formula F6. Part D tests LO4 and Formula F5 directly (mirror the Treynor-Black worked example). Part E tests LO3 — refer back to the Suncor alpha significance discussion.